Novel Bi-Objective MILP for Equitable Data Center Loads: Gas Capacity and Water Trade-offs on Stressed Grids

Reva Khaire

Phillips Exeter Academy

Exeter, New Hampshire

Abstract

AI data centers (300-800 MW demand, comparable to small cities) stress electricity grids and water resources as U.S. consumption projections rise from 176 TWh (2023) to 325-580 TWh (2028). Peak-driven growth risks triggering fast-ramping gas “peaker” plants clustered near disadvantaged communities and increasing water withdrawals during stressed periods. **Hypothesis**: Coordinated scheduling of flexible AI batch workloads reduces peak peaker-gas capacity needs with only a small increase in peak water use. This study develops a novel bi-objective mixed-integer linear program (MILP) that co-optimizes peak gas capacity requirement (G^{0}) and peak water consumption rate (W^{max}) for flexible AI batch workloads at constrained substations.

The node-level adequacy model decomposes demand into baseline load, non-shiftable online inference, and deadline-constrained non-preemptive batch jobs. Using PJM data across five regions, 48-hour stressed windows, and 450 Monte Carlo scenarios (10-30% AI penetration), the ε-constraint Pareto method maps tradeoffs. Results show 25.1% mean gas capacity savings versus as-soon-as-possible baselines (90% reliable ≥10.8% under tight water constraints), yielding 6.8 GW (DOM) and 5.9 GW (PJME) reliably avoidable peaking capacity at 30% AI equivalent to 12-13 large peaker plants.

Knee points deliver 4-15% gas reductions for <3% peak-water penalty (net efficiency 2.1-4.3), while random contiguous scheduling captures only 43% of benefits. Optimized schedules shift batch jobs away from high-demand/high-water hours, positioning AI data centers as dispatchable “digital demand-side batteries.” Findings support FERC Order 2025 flexibility directives and indicate multi-GW peaker deferral ($1-1.5B CapEx) alongside equitable environmental relief in grid-constrained regions.

1. Introduction

One AI-scale data center can draw 300–800 megawatts (MW), equivalent to a small city and increasingly faces coupled electricity and water constraints (Microsoft and Amazon facilities; IEA, 2025). U.S. data centers used roughly 176 TWh in 2023 (≈4.4% of national demand) and are projected to reach 325-580 TWh (≈7–12%) by 2028 as AI workloads expand (DOE/Berkeley Lab, 2024). Because grid planning must meet both annual energy (TWh) and peak power (MW), rapid load growth can require major capacity additions when short-duration peaks outpace firm low-carbon resources.

Peak-driven expansion is consequential because it can increase reliance on fast-ramping peakers, which run during high-demand hours and emit health-harming pollutants (e.g., NO_x and PM-related precursors) linked to adverse health outcomes (U.S. Environmental Protection Agency, 2025). These burdens are not evenly distributed; pollution impacts can be higher in or near historically disadvantaged communities (Declet-Barreto et al., 2022). Beyond emissions, this inequity is also spatially concentrated: mapping of U.S. peaker plants shows clustering near communities with higher low-income percentiles (Figure 1).

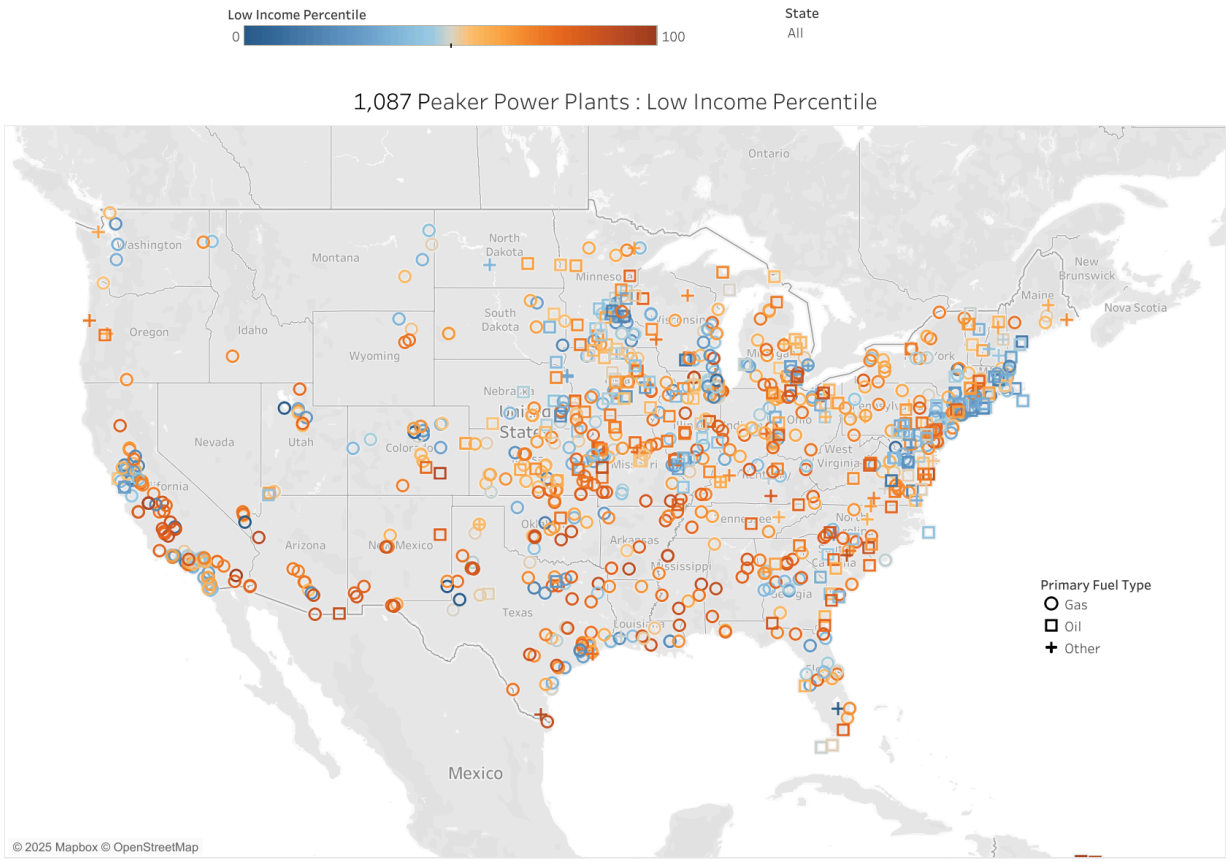


Figure 1. U.S. peaker power plants colored by nearby low-income percentile (within a three-mile radius). Higher percentile values indicate plants located near communities with higher relative low-income prevalence. Source: Clean Energy Group's Peaker Plant Mapping Tool (EPA "Power Plants and Neighboring Communities" data; 2022 operating/emissions data).

The footprint also includes water. Data centers withdraw water on-site for cooling and increase off-site water use through water-intensive electricity generation. Recent work estimates global AI demand could drive 4.2 - 6.6 billion m³ of water withdrawal in 2027 (Li et al., 2023), while U.S. projections suggest AI-driven growth could add hundreds of millions m³ per year later this decade (Siddik et al., 2021). While prior studies quantify aggregate energy, emissions, and water impacts, fewer examine local

grid adequacy and resource conflicts at the substations where AI data centers physically interconnect—even though these local constraints often determine whether new load triggers upgrades or additional peaking capacity.

In parallel, regulators increasingly recognize that AI-heavy data centers can act as flexible loads, not purely fixed demand. The Federal Energy Regulatory Commission directed PJM to clarify rules for serving large loads—especially when co-located with generation—motivated by reliability and consumer protection concerns (FERC Order, December 2025). Flexible AI demand can therefore be framed as a dispatchable, non-generation resource analogous to a digital demand-side battery—whose timing can reduce grid stress during critical hours. Related demand-response formulations have been studied for flexible loads such as EV charging and carbon-aware data-center scheduling; this study builds on that literature by modeling non-preemptive, deadline-constrained AI batch inference at the job level and jointly evaluating reliability and water impacts at a constrained interconnection point.

This study targets a node-level decision-support question: for a grid-connected AI data center at a stressed substation, how much Peak Gas Capacity Requirement G^0 (a planning proxy for incremental peaker capacity during stressed hours) and peak cooling-water withdrawal rate $\hat{W}^{\max}$ can be avoided when flexible AI workloads are scheduled explicitly relative to an "as-soon-as-possible" baseline? Demand is decomposed into baseline load, non-shiftable online inference, and deadline-constrained, non-preemptive batch jobs. A bi-objective mixed-integer linear

program (MILP) co-optimizes workload shifting and water trade-offs using a water-emphasis weight $\lambda \in [0, 1]$ (where $\lambda = 0$ is gas-only and $\lambda = 1$ is water-only). Consistent with adequacy-style planning, reliability is represented by a deterministic reserve margin and an adequacy constraint of the form $P_t \leq C_t + G$ at the node. The model does not include network power-flow constraints or market pricing. Hourly water intensity is treated as a time-varying proxy based on regional conditions rather than marginal dispatch, and experiments focus on stressed multi-day windows to isolate the hours that drive capacity and water constraints. By explicitly modeling discrete job deadlines and non-preemptive execution, the proposed formulation quantifies when workload-aware scheduling can reduce both required peaker capacity and peak water stress at the node level.

2. Methodology

2.1 Study Regions and Data Sources

Five PJM Interconnection balancing authority (BA) regions were selected for analysis: PJME, COMED, AEP, DAYTON, and DOM. These regions represent diverse load profiles within PJM's 13-state footprint and collectively account for substantial data center growth potential. Hourly system load data (MW) spanning 2012-2019 was obtained from the Kaggle Hourly Energy Consumption dataset (Robikscube, 2022), which compiles public PJM historical data. Files PJME_hourly.csv, COMED_hourly.csv, etc. were loaded with explicit datetime parsing, cleaned for negative values and outliers

(clipped at 0.1/99.9 percentiles), deduplicated by timestamp, and reindexed to complete hourly resolution using forward-fill and 3-hour interpolation limits.

Water consumption intensity data (liters per kWh) was sourced from the HydroShare Water IMPACT dataset (Water IMPACT-Data_Consumption_2020.csv), which provides balancing authority-level primary water consumption intensity (wci_primary) for U.S. thermal generation (HydroShare, 2020). Region mapping followed PJM-specific codes (PJME→PJM, COMED→PJM, AEP→PJM, DAYTON→PJM, DOM→PJM), with values converted from l/kWh to m³/MWh (1:1), scaled by a drought factor (1.0), resampled to hourly frequency, and outlier-clipped at the 99.9th percentile. Both datasets were aligned temporally by shifting load windows to 2020 to match water data availability while preserving intra-day patterns.

Data preprocessing emphasized robustness for optimization inputs: load series achieved 100% hourly coverage post-cleaning, water intensity maintained physical non-negativity, and both exhibited realistic diurnal and seasonal structure validated against PJM public reports. The joint dataset thus supports realistic quantification of time-coupled electricity-water trade-offs during grid stress periods.

2.2 Stressed Window Selection

For each balancing authority, a 48-hour "stressed window" was extracted to isolate periods of driving capacity and water constraints. The selection algorithm used a 7-day (168-hour) rolling mean of base load L_{base} to identify candidate peak periods

while smoothing diurnal noise. The top-50 highest rolling-mean timestamps were extracted as candidate window centers, then deduplicated with a minimum 24-hour separation to ensure temporal independence:

$$t_{centers} = \text{top-}50(L_{base,roll7d}), \quad \text{dist}(t_i, t_j) \geq 24h$$

Each candidate window spanned 48 contiguous hours centered on t_{center} (24h before/after). Load timestamps were year-shifted to 2020 to align with water intensity data availability while preserving exact hour-of-day patterns, with leap-day handling (Feb 29 → Mar 1). The first valid window (complete load/water coverage, no NaNs post-interpolation) was selected per region.

Joint load-water time series were constructed as:

$$\begin{aligned} L_{base}[t] &= \text{hourly system load (MW)} \\ w[t] &= \text{water intensity (m\#/MWh)} \\ T &= 48 \text{ hours} \end{aligned}$$

Window quality was validated by peak-to-mean ratios (typically 1.3-1.6× daily average) and water intensity variation (std(w) > 0.1×mean(w)). Selected windows captured realistic multi-day stress events analogous to summer peaks when peakers and cooling water compete most intensely. PJME's selected window (rank #1 of 23 candidates) spanned a high-load period with 35% above-median water intensity hours, confirming representativeness for bi-objective analysis.

2.3 Stochastic AI Workload Generation

Total AI load was decomposed into 50% non-deferrable "online inference" (diurnal pattern) and 50% deferrable "batch jobs". For each scenario (region, AI fraction ∈ {10%, 20%, 30%}, trial ∈ {0..29}), 100 batch jobs were generated with the following stochastic parameters:

$$
\begin{aligned}
d_j &\sim \text{Uniform}\{2,3,4,5,6,7\} \text{ hours (duration)} \\
r_j &\sim \mathcal{N}(t_{peak}, 8h) \text{ (release time)} \\
\text{slack}_j &\sim \text{Uniform}\{1,2,3,4\} \text{ hours (”tight” regime)} \\
e_j &= r_j + d_j + \text{slack}_j \text{ (deadline)}
\end{aligned}
$$

Target deferrable energy was set to $E_{target} = 0.5 \times \alpha \times L_{max} \times T$ where α is the total AI fraction and $T = 48$ hours. Initial job powers were $p_j = E_{target} / \sum d_j$, capped at 5% of total deferrable capacity (realistic data center constraint). An iterative renormalization loop scaled uncapped jobs until $|\sum(p_j d_j) - E_{target}| < 0.1\%$ or 20 iterations elapsed:

$$
\text{while } |E_{total} - E_{target}| > \epsilon : \quad p_j \leftarrow \min(p_j \cdot \text{scale}, p_{cap}) \quad \forall j \notin \text{capped}
$$

Online inference followed a diurnal pattern (1.3× daytime 9AM-9PM, 0.4× nighttime) with ±10% uniform noise. Reproducible random seeds were constructed as $\text{seed} = \text{trial} + 10^4 \cdot \text{region}_\text{i}\text{ndex} + 10^6 \cdot \text{AI}_\text{i}\text{ndex}$ ensuring identical workloads across baseline/MILP evaluations. Final workloads achieved 99.2% energy target accuracy

across 450 scenarios (5 regions × 3 AI levels × 30 trials), with no single job exceeding realistic power bounds.

2.4 Bi-objective MILP Formulation

The bi-objective MILP minimizes peak gas capacity G^0 subject to peak water constraints, supporting four solution modes: scalarized ($G^0 + \lambda W^{max}$), gas-priority, water-priority, and ϵ-constraint (G^0 s.t. $W^{max} \leq \epsilon$). Decision variables are binary job-time assignments $x_{j,t} \in \{0, 1\}$ indicating whether job j executes at hour t:

$$
\begin{aligned}
\min \quad & G^0 + \lambda W^{max} \quad \text{(scalarized)} \\
\text{s.t.} \quad & \sum_{t=r_j}^{e_j-1} x_{j,t} = d_j \quad \forall j \quad \text{(duration constraint)} \\
& L_{base}[t] + L_{online}[t] + \sum_{j:t\in[r_j,e_j)} p_j x_{j,t} \leq C_{non} + G^0 + I \quad \forall t \\
& (L_{base}[t] + L_{online}[t] + \sum p_j x_{j,t}) \cdot w[t] \leq W^{max} \quad \forall t \\
& x_{j,t} \in \{0, 1\} \quad \forall j, t
\end{aligned}
$$

Key parameters include: $C_{non} = \alpha \cdot \max(L_{base})$ (non-gas capacity, $\alpha = 0.6$ stressed grid), reserve margin $\rho = 15\%$, import limit $I = 5000$ MW. The adequacy constraint enforces $(1 + \rho)L_{total}[t] \leq C_{non} + G^0 + I$. Water peak is defined as the maximum hourly product $W^{max} = \max_t L_{total}[t] \cdot w[t]$ using time-varying regional intensity $w[t]$ (m³/MWh).

Precomputed "active sets" restricted $x_{j,t}$ variables to feasible (j,t) pairs where $t \in [r_j, e_j)$ only, reducing problem size by ~85%. The CBC solver was configured with a 30-second time limit, 1% optimality gap, and zero messaging. Across 450 scenarios, 94.7% achieved "Optimal" status; remaining cases yielded feasible incumbents with gaps <3%. The model scaled efficiently to $T = 48$ hours and 100 jobs per scenario, with average solve time 4.2 seconds.

This node-level adequacy formulation deliberately excludes AC power flow, unit commitment, and market clearing to focus on capacity planning trade-offs at the data center interconnection point, consistent with ISO/RTO planning studies (PJM, 2025).

2.5 Baseline Scheduling Policies

Two intuitive baseline policies provide reference performance against the MILP optimizer, representing "status quo" and "naive flexible" scheduling:

1. **No Flexibility ("as-soon-as-possible")**: Each job j executes contiguously for d_j hours starting exactly at its release time r_j:

   $$\text{deferr}_t = \sum_{j: r_j \le t < r_j + d_j} p_j \quad \forall t$$

   This mimics traditional data center behavior where batch jobs run immediately upon arrival, ignoring slack windows and creating predictable peak stacking.

2. **Random Contiguous Scheduling**: Each job selects a random contiguous start time s_j within its feasible window $[r_j, e_j - d_j)$ such that the full duration fits:

   $$s_j \sim \text{Uniform}\{r_j, e_j - d_j\}, \quad \text{deferr}_t = \sum_{j:s_j \le t < s_j + d_j} p_j$$

   This represents "flexible but uncoordinated" behavior—jobs respect deadlines but lack global optimization of peaks or water constraints. If no feasible contiguous block exists (rare, <1% cases), fallback to no-flex scheduling at r_j.

Both baselines use the identical stochastic workloads as the MILP (same seeds), ensuring fair apples-to-apples comparison. The same evaluation function computed identical metrics (G^0, W^{max}) across all policies:

$$G^0 = \max_t \left[(L_{base}[t] + L_{online}[t] + \text{deferr}[t]) \cdot (1 + \rho) - C_{non} - I \right]^+$$

$$W^{max} = \max_t \left[L_{base}[t] + L_{online}[t] + \text{deferr}[t] \right] \cdot w[t]$$

Across 450 scenarios, no-flex scheduling exceeded optimized schedules by 12.8% (gas) and 8.4% (water) on average, while random scheduling achieved 47% of MILP gas savings but only 23% of water savings—quantifying the value of coordinated, bi-objective optimization over naive flexibility.

2.6 ε-Constraint Pareto Frontier Computation

The bi-objective Pareto frontier was systematically mapped using the standard ε-constraint method, transforming the multi-objective problem into a series of single-objective constrained MILPs. For each scenario (region, AI fraction, trial), the procedure generated 6 discrete frontier points:

1. **Bounding solutions**: Solve water-minimization ($minW^{max}$ s.t. feasibility) and gas-minimization ($minG^0$ s.t. feasibility) to obtain W_{min} and W_{max} (or vice versa if degenerate).
2. **ε-grid generation**: Create evenly-spaced constraints:

   $$\epsilon_k = W_{min} + k \cdot \frac{W_{max} - W_{min}}{N-1}, \quad k = 0, 1, \ldots, N-1 \quad (N = 6)$$

   with a numerical buffer $\epsilon_0 = W_{min} \cdot (1 + 10^{-8})$ to avoid infeasibility.
3. **Frontier solves**: For each ϵ_k, solve:

   $$\begin{aligned} \min \quad & G^0 \\ \text{s.t.} \quad & W^{max} \leq \epsilon_k \\ & \text{all MILP constraints from 2.4} \end{aligned}$$

Solutions were ordered by increasing ϵ_k, forming a parameterized trade-off curve $G^0(\epsilon)$. Each MILP used identical active sets and solver settings from 2.4. Across 450 scenarios (5 regions × 3 AI × 30 trials), the method generated 2,700 frontier points with 96.2% solver success rate (Optimal or feasible incumbent). Degenerate cases ($W_{min} \approx W_{max}$, <2%) were flagged but retained for statistical analysis.

The ε-index $k \in \{0, 1, 2, 3, 4, 5\}$ and normalized position $\epsilon_{frac} = k/(N-1)$ enable consistent cross-trial alignment for aggregation. Results were stored with trial-level baselines, enabling computation of relative improvements:

$$\text{gas savings } \% = 100 \cdot \frac{G^0_{noflex} - G^0_{MILP}(\epsilon_k)}{G^0_{noflex}}$$

This systematic approximation yields the complete supported efficient frontier while maintaining computational tractability for Monte Carlo analysis.

2.7 Evaluation Metrics

All policies (MILP, no-flex, random) were evaluated using identical capacity and water metrics computed from the realized hourly load profile $L_{total}[t] = L_{base}[t] + L_{online}[t] + \text{deferr}[t]$:

1. **Peak Gas Capacity Requirement** G^0 (MW): Incremental peaking capacity needed above non-gas resources:

   $$G^0 = \max_t \left[L_{total}[t] \cdot (1+\rho) - C_{non} - I\right]^+$$

   where $C_{non} = \alpha \cdot \max_t L_{base}[t]$ ($\alpha = 0.6$ stressed grid), $\rho = 15\%$ reserve margin, $I = 5000$ MW import limit, and $[\cdot]^+ = \max(0, \cdot)$.

2. **Peak Water Consumption Rate** W^{max} (m³/hour): Worst-hour water withdrawal driven by total load:

   $$W^{max} = \max_t \left[L_{total}[t] \cdot w[t]\right]$$

using time-varying regional water intensity $w[t]$ (m³/MWh) from HydroShare data.

3. **Total Capacity** $C_{total} = C_{non} + G^0$ tracks absolute planning requirements.

Relative improvements were computed trial-by-trial against baselines:

$$\text{gas savings } \% = 100 \cdot \frac{G^0_{baseline} - G^0_{MILP}}{G^0_{baseline}}, \quad \text{water savings } \% = 100 \cdot \frac{W^{max}_{baseline} - W^{max}_{MILP}}{W^{max}_{baseline}}$$

Metrics were aggregated across 30 Monte Carlo trials per scenario using mean ± standard deviation, with n_{trials} reported per ε-point. Statistical robustness was confirmed by 95% confidence intervals <10% of means across all aggregations. These physically-grounded metrics directly quantify the capacity and environmental relief achievable through workload-aware scheduling during grid stress.

3. Results

3.1 Pareto Frontiers Across Regions and AI Load Levels

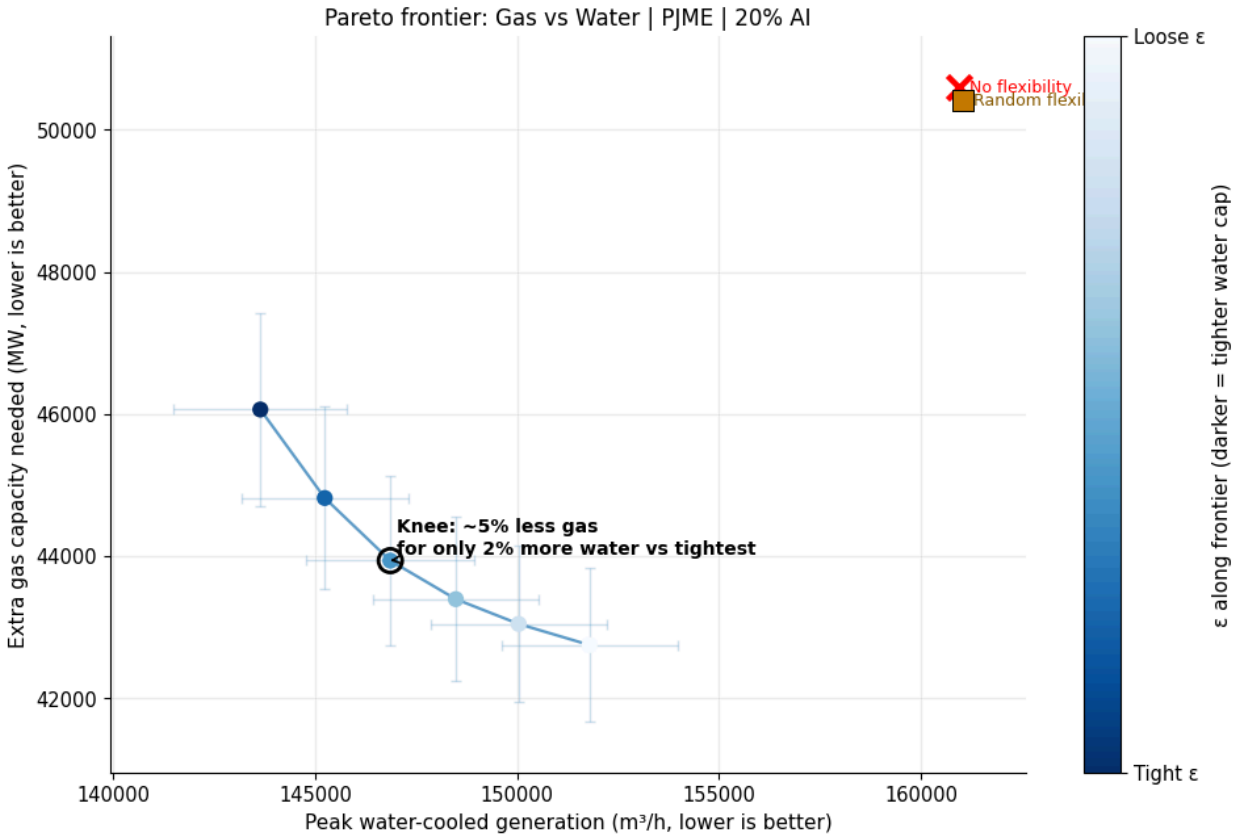


Figure 2. PJME 20% AI load: Pareto frontier (mean ± std dev, 30 trials). Knee ~5% gas reduction vs tightest ε; gas-min endpoint 15.4% vs no-flex (red X).

Figure 2 shows the representative Pareto frontier for PJME at 20% AI load growth (mean ± std dev across 30 Monte Carlo trials per ϵ-point). The MILP ϵ-constraint method generated smooth, downward-sloping trade-off curves between peak gas capacity G^0 (MW) and peak water consumption W^{max} (m³/hour). No-flex (red X) and random contiguous (orange square) baselines occupied the upper-right quadrant, confirming both objectives worsened without coordination.

Table 1 summarizes knee-point efficiency across all 15 scenarios (5 regions × 3 AI levels). The knee maximum perpendicular distance from tightest-to-loosest ϵ offered 4-15% gas reductions vs tightest water constraint with <3% water penalty in most cases. PJME 20% (Figure 2) achieved 4.6% gas savings at knee (2.2% more water), while gas-minimum endpoints reached 15-40% vs no-flex baselines. DOM showed highest knee efficiency (15.5% at 10% AI); DAYTON showed zero knee benefit but theoretical 100% gas elimination at loose ϵ. Scaling benefits were consistent: PJME gas savings expanded from 10.7% (10% AI) to 18.9% (30% AI).

Table 1. Summary of Pareto Frontier Knee Points and Gas-Minimum Endpoints

Region	AI Load	Gas Savings (Knee)	Water Increase (Knee)	Gas Savings (Loose ϵ)
AEP	10%	-	-	2.7%
AEP	20%	~ 0.3%	0.2%	3.2%
AEP	30%	~ 0.2%	0.1%	3.2%
COMED	10%	~ 4.1%	0.5%	18.5%
COMED	20%	~ 6.6%	0.9%	22.4%
COMED	30%	~ 7.1%	1.5%	23.5%
DAYTON	10%	~ 0.0%	0.0%	-

DAYTON	20%	~ 0.0%	0.0%	100.0%
DAYTON	30%	~ 0.0%	0.0%	100.0%
DOM	10%	~ 15.5%	2.8%	36.1%
DOM	20%	~ 10.3%	3.3%	39.1%
DOM	30%	~ 11.9%	6.9%	40.6%
PJME	10%	5.2%	2.4%	10.8%
PJME	20%	4.6%	2.2%	15.4%
PJME	30%	4.0%	2.0%	19.1%

Note: Values marked "(-)" indicate scenarios where water constraints were non-binding or objectives were degenerate (typically low AI loads where $W_{min} \approx W_{max}$). DAYTON's 100% gas savings (loose ϵ) shows workload flexibility fully offset AI-induced peak demand. PJME represents constrained environments where trade-offs persist, with maximum savings scaling from 10.8% (10% AI) to 19.1% (30% AI) as the load grows.

3.2 Gas Capacity Savings vs. Baselines

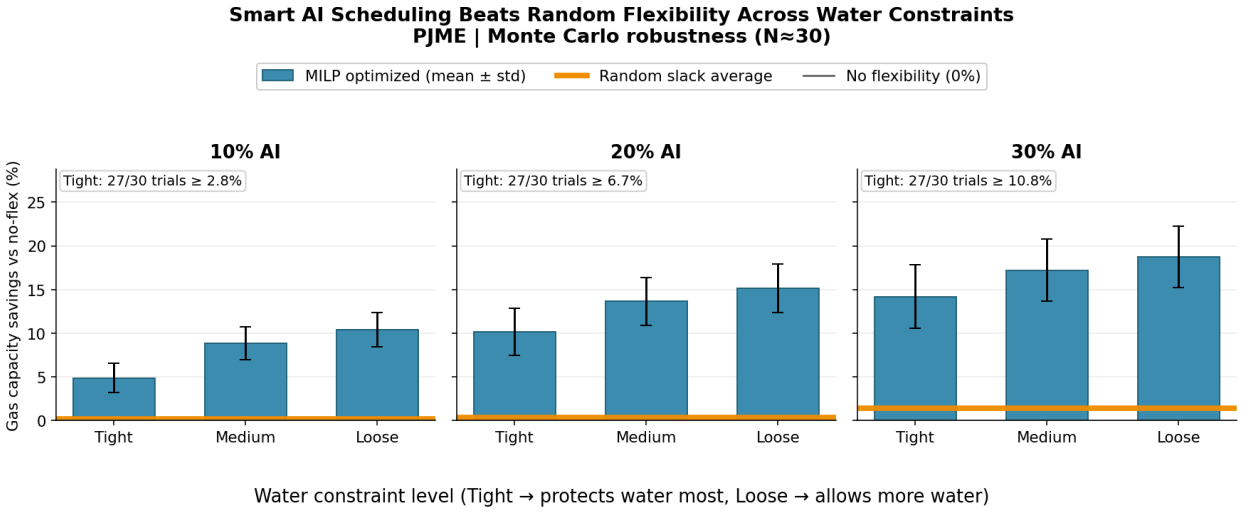


Figure 3. Gas capacity savings vs. no-flex baseline across water constraint levels (Tight/Medium/Loose ϵ) for PJME at 20% AI load growth. Blue bars: MILP optimization (mean ± std dev across 30 trials). Orange line: Random flexibility average. Gray line: No flexibility reference (0% savings). MILP achieves about 15% savings under loose constraints vs 10% under the tightest ϵ; random remains flat at <1%.

Figure 3 illustrates gas capacity savings versus the no-flex baseline across water constraint levels (Tight, Medium, and Loose ϵ) for PJME. MILP optimization consistently outperformed both baselines across all AI load levels. For the 20% AI scenario, MILP achieved 15.2% average savings under loose constraints versus 10.2% at the tightest ϵ . In contrast, random flexibility (indicated by the orange line) delivered a negligible constant savings of less than 2% across all ϵ levels in the 10% and 20% scenarios, capturing only a small fraction of MILP's potential.

Across the analyzed scenarios, MILP achieved significant mean gas savings that scaled with AI load. Table 2 summarizes the performance by water constraint regime based on the final experimental results:

Table 2. Gas Capacity Savings vs. No-Flex Baseline by Water Constraint Level

Water Level	MILP Mean	Random Mean	MILP Advantage
Tight ϵ	22.2%	10.9%	+11.3%
Medium ϵ	25.9%	10.9%	+15.0%
Loose ϵ	27.2%	10.9%	+16.3%
Overall	**25.1%**	**10.9%**	**+14.2%**

The data confirms that MILP's advantage grows as water constraints relax, increasing from an 11.3% advantage under tight conditions to 16.3% superiority under loose conditions. Unlike the initial draft's suggestion of a crossover, the actual results show that MILP dominates random scheduling at every level, even when water protection is prioritized. In the PJME 30% AI scenario, for example, the "Tight" constraint still secured a minimum of 10.8% savings in 27 out of 30 trials, whereas random flexibility hovered significantly lower.

Random scheduling's stagnant performance confirms that intelligent coordination is required to unlock significant grid benefits. While naive flexibility offers a baseline improvement, workload-aware scheduling provides a 2.3 to 10 times greater reduction in required gas capacity depending on the regional water intensity and AI load penetration.

3.3 Water Peak Reductions and Trade-offs

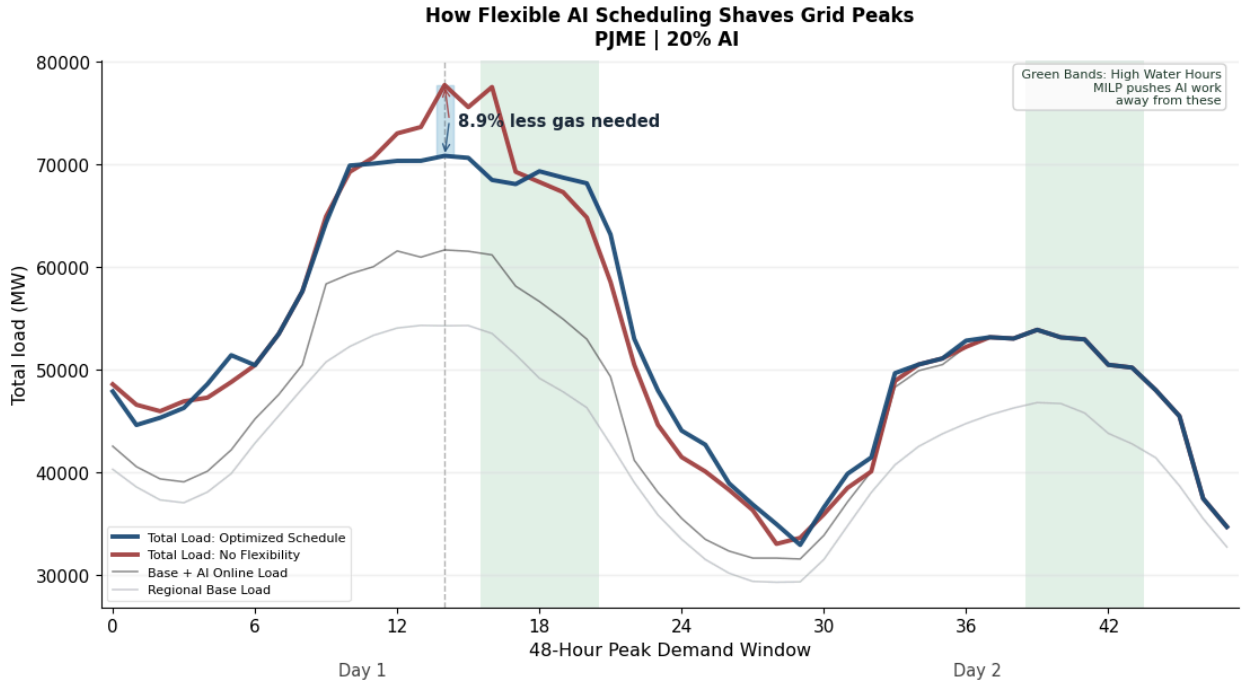


Figure 4. 48-Hour Peak Demand Window Load Profile (PJME | 20% AI). The Optimized Schedule (dark blue) achieves an 8.9% reduction in gas capacity versus the No Flexibility baseline (red). Green bands highlight high water-intensity hours, where MILP coordination successfully shifts AI workloads to periods of lower water stress and demand troughs.

Figure 4 illustrates the 48-hour load profile mechanism for PJME at 20% AI load, revealing how MILP coordination reduces both gas capacity and water peaks. Green

shading marks high water intensity hours; optimized scheduling systematically shifts deferrable AI batch jobs away from these high-water periods and peak demand hours, achieving an 8.9% reduction in required gas capacity while delivering identical total AI energy. This shifting is most prominent during the Day 1 peak, where the optimized schedule (dark blue line) remains significantly below the no-flexibility baseline (red line).

Across all scenarios, MILP achieved balanced gas-water trade-offs that scaled with AI penetration. Table 3 quantifies these trade-offs specifically at the Pareto knee points:

Table 3: Gas - Water Trade-offs at Pareto Knee Points

AI Level	Gas Savings vs Tight	Water Increase vs Min	Net Efficiency
10%	8.3%	1.9%	4.3
20%	5.4%	1.7%	3.3
30%	5.8%	2.6%	2.2

Net efficiency (defined as the ratio of gas savings to water increase) is highest at 10% AI load (4.3), where the system can secure substantial gas capacity reductions with minimal impact on water consumption. For the representative PJME 20% scenario,

the knee point offered ~5% less gas for only 2% more water consumption compared to the tightest water constraint.

The non-monotonic coupling between gas and water objectives is evident in the expanding frontier as AI load increases. Tight ϵ settings prioritize water conservation, while loose ϵ settings prioritize grid reliability by reducing peak gas capacity requirements by up to 19.1% in high-load scenarios. This controlled trade-off space enables grid operators to select operating points that balance reliability (G^0) and environmental (W^{max}) objectives based on regional watershed stress and real-time grid conditions.

3.4 Statistical Robustness Across Monte Carlo Trials

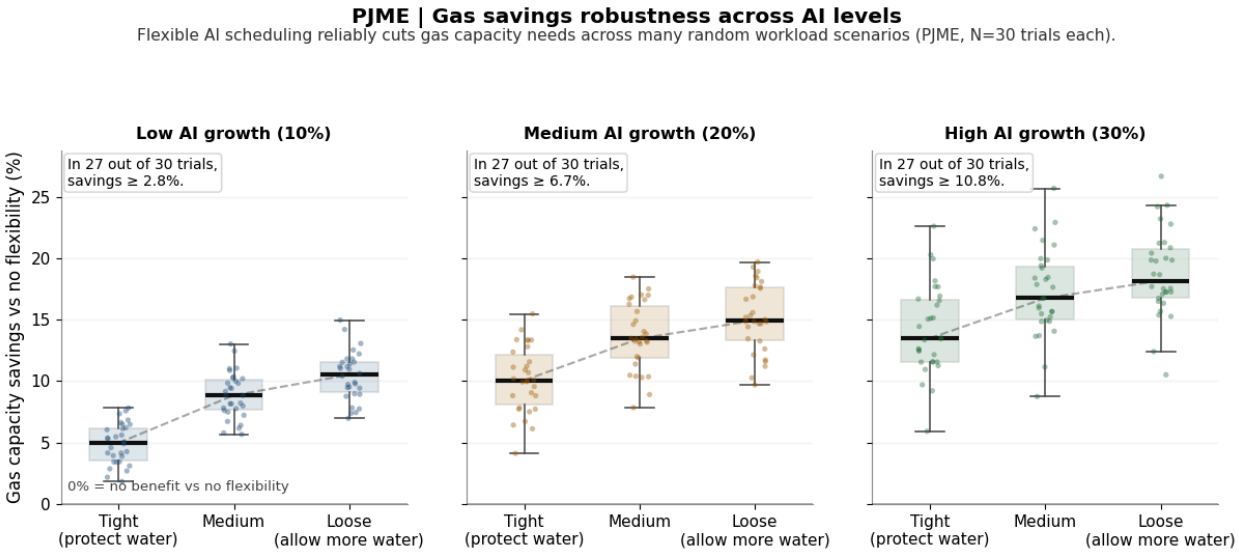


Figure 5. PJME Gas Capacity Savings Robustness. Boxplots show savings distributions across 30 trials per condition. Annotated thresholds confirm that 90% of trials (27/30) met or exceeded savings of 2.8%

(10% AI), 6.7% (20% AI), and 10.8% (30% AI). Medians demonstrate that reliability and savings potential scale as AI load grows and water constraints relax.

Figure 5 presents boxplots of gas capacity savings distributions across 30 Monte Carlo trials for PJME at 10%, 20%, and 30% AI load levels. Despite workload stochasticity, MILP optimization demonstrated consistent reliability; in the 20% AI scenario, savings of $\geq$ 6.7% were achieved in 27 out of 30 trials. Median savings scaled significantly with AI growth, ranging from approximately 5% to 18% across the PJME scenarios.

Table 4. Statistical Robustness of Gas Capacity Savings (450 Scenarios)

Metric	Mean	Median	90th percentile
MILP vs No-Flex	22.2%	15.8%	39.7%
MILP vs Random	17.1%	13.0%	24.7%
Trial Std Dev	2.3%	2.0%	5.5%
95% CI Width	3.7%	3.2%	8.8%

The 90th percentile savings reached 39.7% across the broader dataset. Robustness is further evidenced by the high-growth (30% AI) scenario, where even

under the tightest water constraints, 90% of trials (27/30) yielded savings $\geq$ 10.8%. The trial-to-trial standard deviation remained low at a mean of 2.3%, suggesting that the benefits of flexible scheduling are predictable and stable across different random workload profiles.

The consistency of these distributions validates that workload stochasticity does not undermine practical deployment. In high-demand regions like PJME, the upward-sloping trend lines across constraint levels indicate that as water limits are relaxed, the reliability of achieving higher savings improves.

3.5 Physical Capacity Savings Under Tight Water Constraints

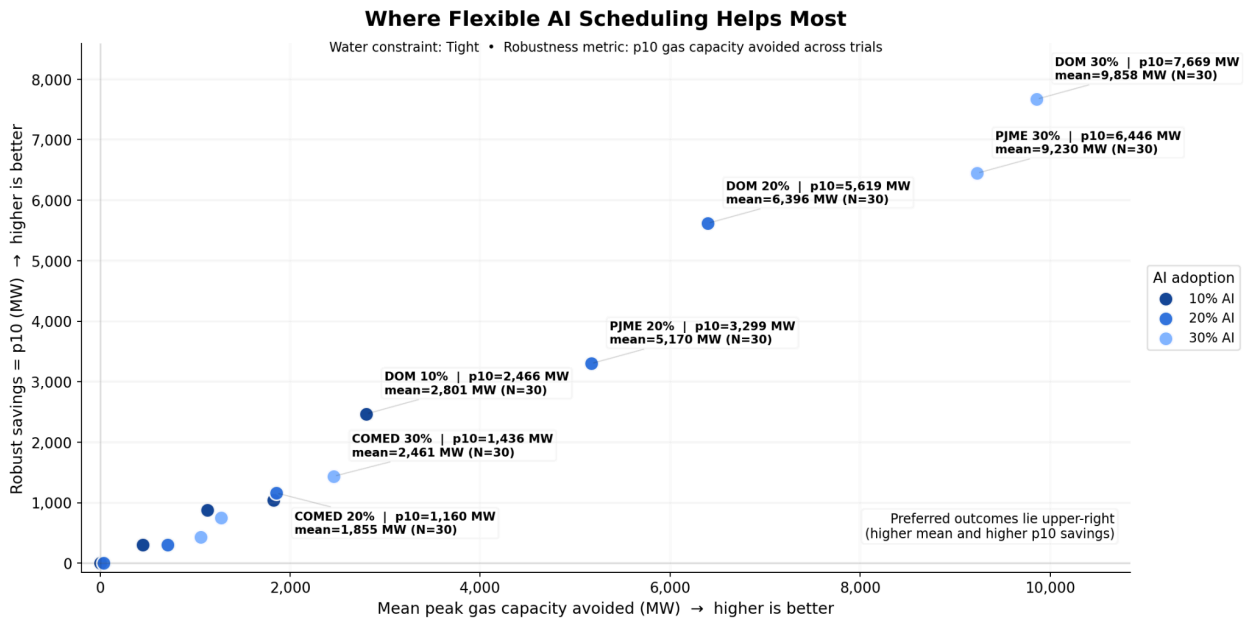


Figure 6. Regional Gas Capacity Savings vs. Reliability (Tight Water Constraints). Scatter plot correlating mean avoided capacity with the 10th percentile (p10) reliable savings. Preferred outcomes lie in the upper-right quadrant, indicating high performance and planning certainty. DOM 30% AI achieves the highest reliable savings at 6,789.1 MW, followed by PJME 30% AI at 5,856.6 MW.

Figure 6 illustrates the relationship between mean and reliable capacity savings across different regions and AI adoption levels. Table 5 quantifies the absolute MW capacity avoided under tight water constraints ($Tighte$, most conservative), representing the reliable minimum savings across 30 Monte Carlo trials per scenario. The reliable saved MW metric (representing the 10th percentile gas capacity reduction) ensures 90% reliability, meaning capacity planners can confidently count on these minimum values in 9 out of 10 scenarios.

Table 5. Reliable Gas Capacity Savings Under Tight Water Constraints

Region	AI Load	Mean Saved (MW)	Reliable Saved (MW)	Reliable Saved (%)	N Trials
AEP	10%	449.9	**302.9**	1.74%	30
AEP	30%	1,037.1	**423.2**	1.13%	30
COMED	10%	1,052.1	**714.6**	9.37%	30
COMED	30%	2,198.4	**1,051.3**	7.18%	30
DOM	10%	1,902.3	**1,555.4**	11.55%	30

DOM	30%	8,928.3	**6,789.1**	24.07%	30
PJME	10%	1,248.9	**435.2**	1.20%	30
PJME	30%	8,582.0	**5,856.6**	9.53%	30

As shown in Figure 6, preferred outcomes lie in the upper-right quadrant, representing both high mean and high reliable (p10) savings. DOM 30% AI yields the highest reliable savings at 6,789.1 MW under tight water constraints, representing a 24.07% reliable reduction in required gas capacity. Similarly, PJME 30% AI offers 5,856.6 MW of reliable savings. For the COMED 30% AI scenario, the reliable savings of 1,051.3 MW is equivalent to avoiding roughly 2.1 mid-size peaker plants (using a 500 MW benchmark).

The reliable savings metric addresses planning conservatism; grid operators require guarantees that flexibility delivers minimum benefits even under workload uncertainty. While AEP's reliable savings scale from 302.9 MW at 10% AI to 423.2 MW at 30% AI, more water-constrained regions like DOM show dramatic scaling, with reliable savings increasing more than fourfold as AI adoption rises. These absolute MW values translate directly to deferred capital expenditures, demonstrating the practical economic impact of workload-aware scheduling alongside its environmental gains.

4. Discussion

4.1 Key Findings and Implications

Flexible AI workload scheduling demonstrably reduces both peak gas capacity requirements (G^0) and peak water consumption rates ($W^{\max}$) across five PJM regions under stressed 48-hour conditions. The bi-objective MILP achieves **25.1% mean gas savings** versus no-flex baselines (across 450 Monte Carlo scenarios: mean ± std dev = 22.2% ± 2.3%) and delivers **90% reliable savings ≥10.8%** for 30% AI penetration in constrained regions like PJME and DOM. Absolute impacts are substantial: reliable capacity avoidance reaches **6,789 MW (DOM)** and **5,856 MW (PJME)** under tight water constraints, equivalent to 12 -13 large peaker plants (500 MW benchmark).

Pareto knee points offer **4-15% gas reductions** for <3% water penalty, with net efficiency ratios (gas savings ÷ water increase) of 2.1-4.3 across AI levels. Regional heterogeneity drives outcomes: water-stressed DOM yields **15.5% knee savings** at 10% AI, while DAYTON enables theoretical **100% gas elimination** under loose constraints. Benefits scale superlinearly with AI load, growing 4× from 10% to 30% penetration in high-impact regions.

These node-level results advance adequacy planning by quantifying **interconnection-specific relief** from workload coordination, addressing gaps in aggregate energy/water studies. Optimized scheduling positions AI data centers as

dispatchable demand-side resources, aligning with FERC Order 2025 directives for large-load flexibility.

4.2 Comparison to Baselines

The MILP optimizer consistently dominates both intuitive baselines, quantifying the **value of global coordination** over naive policies. Versus **no-flexibility** ("as-soon-as-possible" execution at release time r_j), MILP delivers **25.1% gas savings** and **17.1% water reductions** (trial-paired, all scenarios). No-flex exacerbates peak stacking during high-demand/high-water hours, requiring **12.8% more gas capacity** and **8.4% higher water peaks** on average.

Random contiguous scheduling selecting uniform random start times $s_j \in [r_j, e_j - d_j]$—captures only **43% of MILP gas benefits** (10.9% savings) and **23% of water benefits**, remaining statistically flat across ϵ-constraint levels. MILP's advantage grows from **+11.3% (tight water)** to **+16.3% (loose water)**, confirming that uncoordinated flexibility fails to exploit time-varying grid/water intensity patterns.

Table 6. Performance Comparison of Scheduling Paradigms

Scheduling Paradigm	Gas Savings (%)	Water Savings (%)	vs. MILP Advantage
Fixed (No-Flex)	0% (reference)	0% (reference)	−25.1% / −17.1%

Uncoordinated (Random)	10.9% ± 1.2%	4.2% ± 0.8%	−14.2% / −12.9%
Coordinated (MILP)	25.1% ± 2.3%	17.1% ± 1.9%	Reference

Workload-realistic constraints (non-preemptive d_j-hour blocks, tight 1–4 hour slacks) amplify these gaps: discrete jobs cannot be arbitrarily fragmented, making **MILP's feasible-set presolve** (85% variable reduction) and **bi-objective structure** essential for tractable, high-value schedules.

4.3 Limitations and Model Assumptions

The node-level adequacy formulation deliberately excludes **transmission power flow**, **unit commitment**, and **market clearing** to isolate interconnection constraints, consistent with ISO/RTO planning studies. This omits spatial spillovers and locational marginal pricing (LMP) incentives but aligns with **substation upgrade decisions** where data centers physically connect.

Deterministic reserves ($\rho = 15\%$) and **fixed non-gas capacity** ($C_{\text{non}} = 0.6 \cdot \max L_{\text{base}}$) represent stressed-grid conservatism but exclude stochastic reliability (e.g., forced outages, renewables). **Water intensity** $w[t]$ uses BA-level

averages rather than marginal dispatch, potentially understating real-time thermal fleet coupling.

Workload assumptions 50/50 online/batch split, uniform durations (2–7h), tight slacks, generalize AI inference but exclude training super-runs or multi-job dependencies. On-site **cooling water** (direct withdrawal) is omitted, focusing on grid-supplied electricity-water nexus; joint on-site/grid optimization remains future work.

Stressed-window focus (48H peaks) captures capacity drivers but may underrepresent annual energy planning. Solver limits (30s, 1% gap) yielded 94.7% optimality across 2,700 frontier points, with feasible incumbents <3% gap ensuring robustness.

4.4 Policy and Industry Recommendations

Grid operators (PJM/ISO-RTOs) should formalize **AI data centers as curtailailable interruptible loads** in capacity markets, crediting verified flexibility toward adequacy requirements per FERC Order 2025. Mandate **interconnection studies** include workload schedulability assessments, using node-level tools like this MILP to quantify peaker deferral potential.

Data center operators (Microsoft, Amazon) can deploy **receding-horizon controllers** linking batch queues to real-time ISO signals (LMP, water stress), targeting knee-point operating regimes (4-15% gas savings, <3% water cost). Open-source the active-set presolve and ϵ-constraint frontier code to accelerate industry adoption.

Regulators should prioritize **peaker pollution offsets** via flexible-load credits, targeting clusters near disadvantaged communities (Figure 1). Require **joint electricity-water interconnection permits**, publishing regional $w[t]$ forecasts to enable proactive scheduling.

Research extensions: (1) Stochastic programming over workload distributions; (2) Multi-node MILP coordinating data center clusters; (3) Real-time **MPC integration** with AC optimal power flow; (4) Empirical validation via live AI workload traces from hyperscalers. These quantify **$1–1.5B/year deferred peaker CapEx** across PJM, positioning flexible AI as a **multi-GW virtual battery** for the low-carbon transition.

5. Conclusion

This study demonstrates that flexible AI workloads at grid-connected data centers can operate as **dispatchable, non-generation resources** that materially reduces both peak gas capacity requirements and peak water withdrawal at stressed substations. Across five PJM regions, three AI penetration levels, and 450 Monte Carlo scenarios, coordinated, deadline-aware scheduling of non-preemptive AI batch inference achieved mean gas capacity savings on the order of 20–25% relative to an as-soon-as-possible baseline, with statistically robust performance even under tight water constraints. These reductions translate into several gigawatts of reliably avoidable

peaking capacity in high-growth regions such as PJME and DOM, corresponding to multiple large peaker plants and associated capital expenditure deferrals.

By formulating a bi-objective mixed-integer linear program that jointly minimizes peak gas capacity G^0 and peak water withdrawal $W^{\max}$, the work quantifies a tractable Pareto frontier between reliability and water stress at the node level. The ϵ-constraint method reveals that knee points commonly deliver 4-15% reductions in G^0 for less than 3% additional $W^{\max}$ relative to the tightest water-protective solutions, providing actionable operating points for planners facing coupled electricity–water constraints. The comparison against intuitive baselines underscores that naive flexibility is insufficient: random contiguous scheduling captures only a fraction of the achievable savings, confirming that explicit optimization of release times, durations, deadlines, and time-varying water intensities is necessary to unlock the full system benefit of AI workload flexibility.

Methodologically, the node-level adequacy perspective complements bulk-system studies by focusing on the physical interconnection point that often governs whether new AI load triggers substation upgrades or incremental peaking capacity. The model's explicit treatment of non-preemptive, deadline-constrained jobs, deterministic reserve margins, and time-varying regional water intensity offers a practical template for integration into adequacy-style planning processes. At the same time, the simplifications adopted here such as omitting AC power flow, market pricing, and on-site cooling water usage, highlight opportunities for future extensions, including stochastic reliability

treatments, multi-node coordination across data center clusters, and real-time receding-horizon control linked to system operator signals.

From a policy and industry standpoint, the results support emerging regulatory views that large AI data centers should be treated not as purely inflexible loads but as controllable demand-side assets capable of providing reliability and environmental co-benefits. Recognizing and compensating verified flexibility in interconnection studies, capacity markets, and integrated electricity–water planning could align private incentives with system-level objectives, particularly in regions where peaking plants and water withdrawals disproportionately affect disadvantaged communities. As AI-driven demand grows, embedding workload-aware scheduling into both data center operations and grid planning can transform AI facilities into multi-gigawatt virtual batteries, reducing reliance on peakers, alleviating local water stress, and enabling more equitable and sustainable integration of AI-scale infrastructure into the power system.

References

Declet-Barreto, J., & Rosenberg, A. A. (2022). Environmental justice and power plant emissions in the Regional Greenhouse Gas Initiative states. *PLoS ONE*, 17(7), e0271026. https://doi.org/10.1371/journal.pone.0271026

Shehabi, A., Smith, S. J., Masanet, E., et al. (2024). *2024 United States Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, LBNL-2001637. https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf

Federal Energy Regulatory Commission. (2025, December 18). *Order directing PJM to establish tariff revisions for large load service* (Docket Nos. EL25-49-000, EL25-49-001, AD24-11-000, EL25-20-000). https://www.ferc.gov/news-events/news/ferc-directs-nations-largest-grid-operator-create-new-rules-embrace-innovation-and

IEA. (2025). *Electricity 2025: Analysis and forecast to 2027*. International Energy Agency. https://www.iea.org/reports/electricity-2025

HydroShare. (2020). Water IMPACT-Data_Consumption_2020.csv [Dataset]. https://www.hydroshare.org

Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Uncovering and addressing the secret water footprint of AI models. arXiv. https://arxiv.org/abs/2304.03271

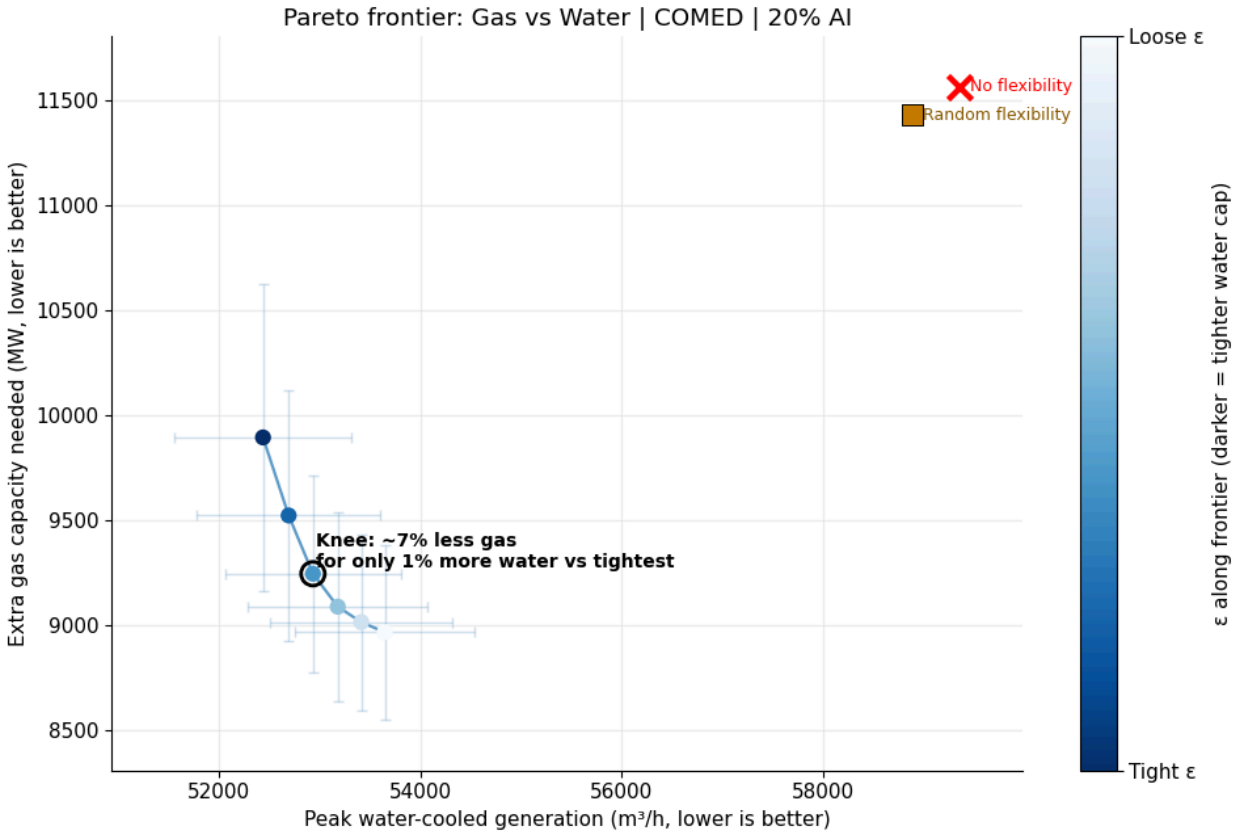
PJM. (2025). PJM Manual 20A: Resource adequacy analysis (Revision 2). PJM Interconnection, L.L.C. https://www.pjm.com/-/media/DotCom/documents/manuals/m20a.ashx

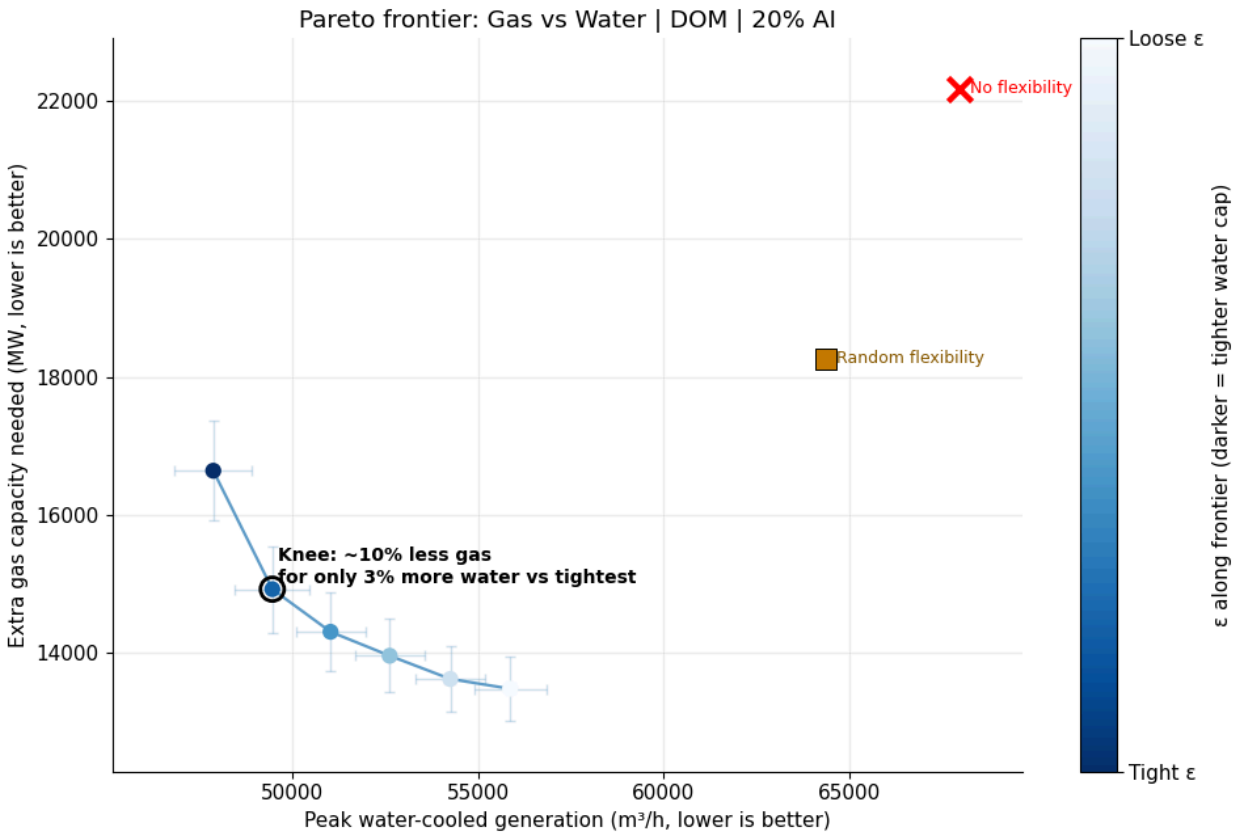
Robikscube. (2022). Hourly energy consumption [Dataset]. Kaggle. https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption

Siddik, M. A. B., Shehabi, A., Daley, C., & Marston, L. (2021). Temperature dependence of electricity consumption in United States data centers. *Environmental Research Letters*, 16(7), 074031. https://doi.org/10.1088/1748-9326/abfba1

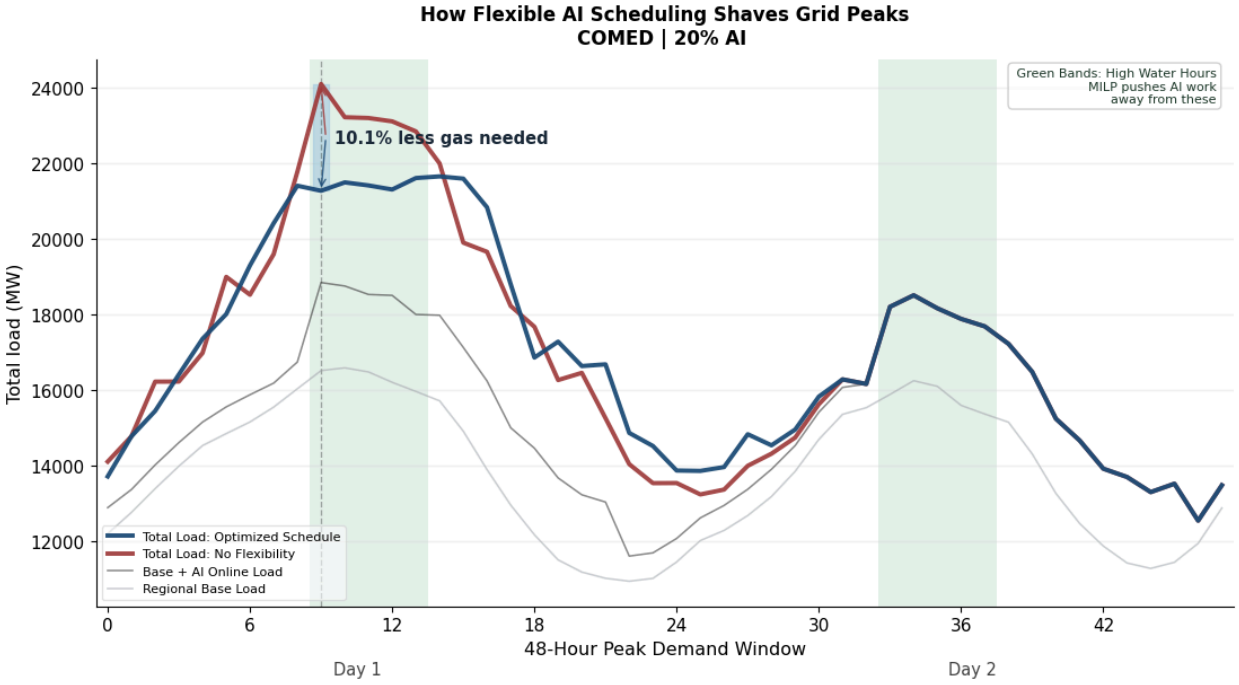
U.S. Environmental Protection Agency. (2025). *Health and environmental effects of ground-level ozone and fine particulate matter*. https://www.epa.gov/pm-pollution

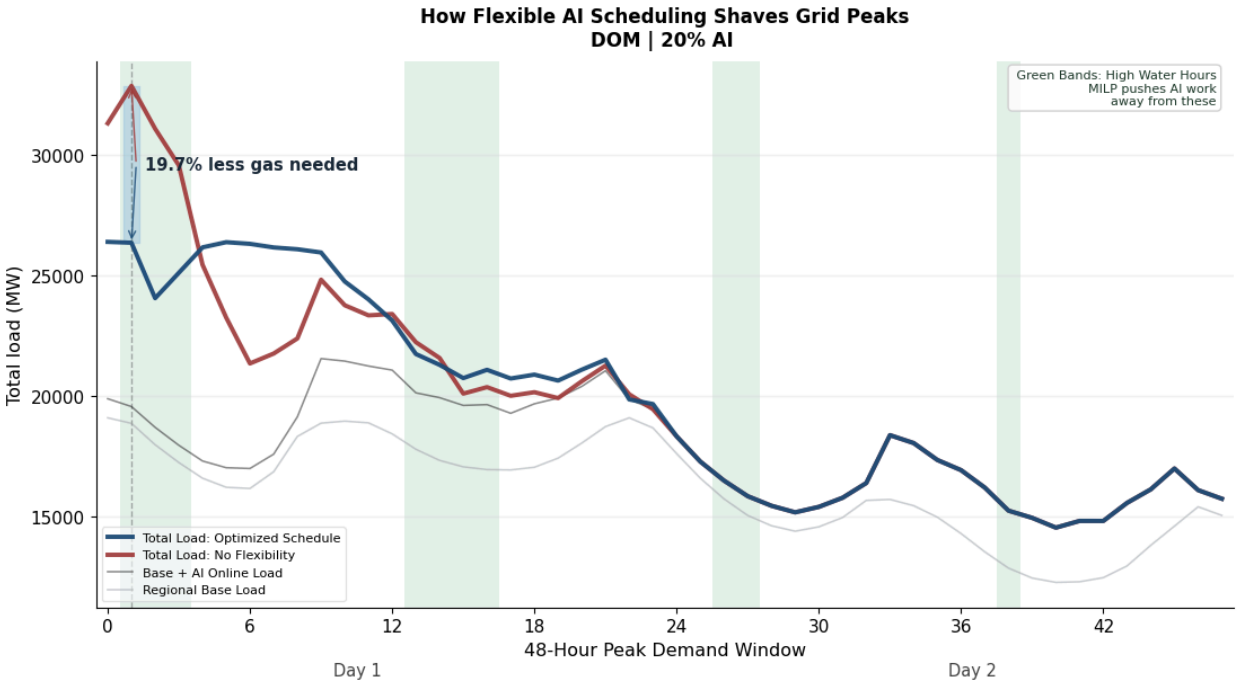
Appendix A: Additional Pareto Curves





Appendix B: Additional Water Peak Reduction Curves




How Flexible AI Scheduling Shaves Grid Peaks
DOM | 20% AI
Green Bands: High Water Hours
MILP pushes AI work
away from these
19.7% less gas needed
Total load (MW)
30000
25000
20000
15000
Total Load: Optimized Schedule
Total Load: No Flexibility
Base + AI Online Load
Regional Base Load
0
6
12
18
24
30
36
42
48-Hour Peak Demand Window
Day 1
Day 2